Yaowenqi (Howard) Liu

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RESEARCH INTERESTS

Reliable LLM Agent: Making LLM agents reliable. I study this with verifiable feedback such as retrieved evidence, uncertainty estimates, formal languages, and reward-guided training.

Formal Reasoning: Using formal languages such as Lean 4 to encode task rules and constraints, and to provide verifiable feedback on LLM reasoning, actions, and outcomes.

AI4Sci: Agents for automated research in machine learning: empirical work grounded in code execution and retrieved literature, and theoretical work checked with Lean 4.

EDUCATION

University of Illinois Urbana-Champaign Jan 2026 – Dec 2026
Master of Computer Science GPA: 3.9 / 4.0

University of Illinois Urbana-Champaign Aug 2022 – May 2025
B.S. in Computer Science & Statistics, Minor in Mathematics GPA: 3.9 / 4.0

Relevant Coursework:
Machine Learning: Machine Learning (A), Reinforcement Learning (A), Statistical Learning (A)
NLP & Information Retrieval: Natural Language Processing (A+), Advanced Information Retrieval (A), Data Mining (A)
Computer Vision: Deep Learning for Computer Vision (A), Computational Photography (A+)
Mathematics & Statistics: Numerical Methods (A), Statistics and Probability I (A+), Statistics and Probability II (A), Linear Programming (A), Nonlinear Programming (A), Stochastic Processes (A+)

PUBLICATIONS

* Co-first author.

MLP-Bench: Towards End-to-End Formal Research Assistance in Machine Learning Theory
Yaowenqi Liu*, Ruida Wang*, Rui Pan*, Yuxing Liu, Yifan Hao, Tong Zhang.
Preprint, 2026. [PDF]

GUIDE: Towards Scalable Advising for Research Ideas
Yaowenqi Liu*, Bingxu Meng*, Rui Pan*, Yuxing Liu, Jerry Huang, Jiaxuan You, Tong Zhang.
ACL 2026 Main. [ACL Anthology]

Optimal Aggregation of LLM and PRM Signals for Efficient Test-Time Scaling
Peng Kuang, Yanli Wang, Xiaoyu Han, **Yaowenqi Liu**, Kaidi Xu, Haohan Wang.
ICLR 2026. [arXiv]

AgentSPEX: An Agent SPecification and EXecution Language
Pengcheng Wang*, Jerry Huang*, Jiarui Yao*, Rui Pan, Peizhi Niu, **Yaowenqi Liu**, Ruida Wang, Renhao Lu, Yuwei Guo, Tong Zhang.
Preprint, 2026. [arXiv]

RESEARCH EXPERIENCE

Contributor, ConnectBench Oct 2026 – Present
UIUC × AWS

- Developing the formalization module in Lean to encode customer-service policies, access permissions, and database constraints for verifying LLM agents' actions and final outcomes.

Research Intern, Prof. Tong Zhang's Group, UIUC Jan 2025 – Present

- **GUIDE (ACL 2026 Main, first author):** Developed a retrieval-augmented reviewer over 200K ML papers and an RLHF training framework to evaluate research novelty and significance.
- **MLP-Bench (Preprint, first author):** Led a 360-problem benchmark for ML theory research assistance; formalized 120 theorem statements and constructed machine-checked proofs in Lean 4.
- **AgentSPEX (Preprint):** Integrated MCP-enabled tools for the GUIDE reviewer demo and implemented a chunk-based reader to reduce latency when processing long documents.

ACADEMIC SERVICE

Reviewer: ICLR 2027; NeurIPS 2026 Workshop on GenAI4Health.